GG-C-00716C(DSA-DM)
28 March 1974
SUPERSEDING
Int. Fed. Spec. GG-C-00716B(DSA-DM)
19 December 1967 and
Interim revision of
Fed. Spec. GG-C-716
September 2, 1942

INTERIM FEDERAL SPECIFICATION

CRUTCH, ADJUSTABLE, WOOD

This Interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this Interim Federal specification as a valid exception to Federal Specification GG-C-716.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers adjustable wooden crutches (see 6.1).

1.2 Classification. The crutches covered by this specification shall be of the following types, as specified (see 6.2):

Type I. Solid wood.
Type II. Laminated wood (Bows).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636 Box, Fiberboard.
PPP-B-640 Boxes, Fiberboard, Corrugated, Triple-wall.

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STANDARDS

Federal

FED-STD-123 Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

SPECIFICATIONS

Military

MIL-W-6100 Wood, Determination of Moisture Content of.

STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center.

1781 Crutches, Adjustable.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 OTHER PUBLICATIONS. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

American Society for Testing and Materials (ASTM) Standard

D805-63 Testing Veneer, Plywood, and Other Glued Veneer Constructions.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Materials. Materials used in the manufacture of crutches shall meet the requirements specified herein, and as shown on the drawing. When materials to be used are not specifically designated, those used shall be of the highest commercial quality and entirely suitable for the intended purpose.

3.1.1 Wood. Wooden parts of the crutches shall be manufactured of solid wood, for type I, as specified in 3.1.1.1; or laminated wood, for type II, as specified in 3.1.1.2.

3.1.1.1 Type I, Solid wood. Solid wood shall be hickory, commercial white ash, maple, birch, or oak. The average number of annual rings of ring porous woods, if used, shall not exceed 15 per inch of radius. Wood shall be sound, thoroughly seasoned (12 to 18 percent moisture); shall be tested as specified in 4.4.2; and shall be free from knots, splits, decay, bird pecks, checks, and any other defects affecting its appearance or which may affect serviceability. Solid wood shall be commercial straight grain, which as used herein shall mean; the grain of the wood shall not cross from one side of the piece to the other side, in less than eight inches of length.

3.1.1.2 Type II, Laminated wood. Laminated wood shall be cut from panels made up from five, seven, or nine plies of veneer as specified herein. This shall be the type of plywood known commercially as parallel grain construction. The resulting bonded panels shall have a dry shear of not less than 1200 pounds per square inch, with a minimum average of not less than 1700 pounds. Wet shear shall not be less than 850 pounds per square inch, with a minimum average of not less than 1200 pounds per square inch. Testing shall be done as specified in 4.4.3. Not less than 10 determinations for each type of shear shall be made from each lot of laminated wood to determine the above averages. There shall be no signs of delamination after soaking in water and prior to testing.

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3.1.1.2.1 Face veneers. The face veneers (face and back) shall be hickory, ash, maple, or yellow birch calipering a minimum of 0.039 inch and maximum of 0.050 inch prior to sanding, except for five ply laminated bows each veneer shall be 0.100 ± 0.003 inch thick, after sanding and prior to gluing. The veneer shall be straight grained and free from any evidence of decay. There shall be no knots, splits, wormholes, bird pecks, or checks evident in the face veneers. The veneer shall have a moisture content of not more than 10 percent at the time of gluing. Moisture content shall be tested as specified in 4.4.2.

3.1.1.2.2 Inner Veneers. The inner veneers shall be yellow birch, maple, hickory, or ash. The veneers used shall caliper between 0.060 and 0.130 inch except the five ply laminated veneers shall each be 0.100 ± 0.003 inch thick, after sanding, and prior to gluing. Final thickness shall be governed by bow and assembly requirements (see 3.1.2 and 3.2.2). The veneers shall not exceed 10 percent moisture at the time of gluing. Moisture content shall be tested as specified in 4.4.2. Inner veneers shall be glued in the same manner as the face veneers, and in each case, the individual finished sheet of veneer shall be made of components of a single species. When the inner veneers are made up of several pieces of veneer in lieu of a single one piece sheet, the components shall be tapeless spliced, using veneer of one species bonded together on a tapeless splicer. Quality of the inner veneers shall be the same as the face veneers, except small firm knots shall be allowed. There shall be no voids.

3.1.2 Bolts. There shall be three 10/24 commercial stove bolts, 3/16 inch in diameter. Bolts shall be the lengths shown on the drawing. Bolts shall have a head measuring 1/2 inch diameter plus zero, minus 1/16 inch, or a washer conforming to these dimensions shall be used. The same size washer shall be used under the wing nut. A self-locking wing nut shall be used. The wing nut shall be Amerace Esna Corporation's ESNA PART NO. WC-04, or equal. Bolts and washers shall be steel, cadmium or zinc plated. The wing nut shall be zinc base, die cast alloy. Metal parts shall be finished smooth and free of burrs and sharp edges. Wing nuts and washers shall provide a proper fit on the bolt. Bolts shall completely engage wing nuts and protrude a minimum of one screw length (one complete turn) beyond the plastic insert when the crutch is assembled and nuts are hand tightened. The bolt shall not protrude beyond the height of the wing nut.

All bolts shall be given a suitable treatment for relief of hydrogen embrittlement after electroplating.

3.2 Construction. Each crutch shall consist of the arm or supporting piece, two bows, hand piece, foot piece, and three stove bolts, fitted with washers and winged nuts. The complete crutch shall be capable of adjustment from 45-1/4 inches to 57-1/4 inches in overall length, when measured from the bottom of foot piece to the top central portion of the armpiece midway between the bows. The sum of the allowed tolerances shall not increase or decrease the overall length more than 1/8 inch. The assembled crutches and component parts shall be as shown in DPSC Drawing 1781.

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3.2.1 Armpiece. The armpiece shall be 1-1/2 inches ± 1/4 inch in thickness (depth); 7-3/4 inches ± 1/8 inch in length; and not less than 7/8 inch in width. The upper surface shall form the concave arc of a circle having a radius of eight inches. The lower surface shall have two mortises, 4, + 1/2-0 inches between centers, spaced equidistant from sides and ends, to receive the tenons of the upper ends of the bows. The edges and corners shall be rounded. As an alternate, the bottom side of the armpiece shall have a concave curve between the mortises, however, the armpiece must pass the strength test specified in 4.4.1. As an alternate, the armpiece shall be 13/16 + 1/16, -0 inch wide.

3.2.2 Bows. The bows shall be fabricated from wood as specified in 3.2.1.1 or 3.2.1.2. If supplied with wood as specified in 3.2.1.2 each face veneer shall be a single, one piece veneer. The bows shall be 7/8 inch by 1/2 inch, and 41-1/2 inches ± 1/8 inch in length, with a square or round tenon, not less than 1/2 inch in length on the upper end. The assembled crutch shall have the wider surfaces opposite each other. Each bow shall have five holes as shown on the drawing, drilled through the center of the wider surfaces, for attaching the handpiece. Two holes shall be drilled in a similar manner at a point 1-1/2 inches and 6-1/2 inches from the lower end, for attachment of the footpiece. Holes shall be proper clearance size for 3/16 inch stove bolts. The holes shall be drilled accurately so that they will line up properly, for easy assembly of handpiece and footpiece to bows.

3.2.3 Handpiece. The handpiece shall be cylindrical in shape, 4 inches in length and 1-1/8 inches in diameter at the center, tapering toward each end to 3/4 inch in diameter. There shall be a hole drilled through the center from end to end which shall be proper clearance size for the 3/16 inch stove bolts.

3.2.4 Footpiece. When laminated bows are used, the footpiece shall be one piece, 13/16 inch minimum to 1-1/8 inch maximum, square or rectangular; and 22 inches ± 1/8 inch in length. When solid wood bows are used, the footpiece shall be 1-1/4 inches wide by 7/8 inches thick, and 22 inches ± 1/8 inch in length. Drill holes of the latter shall be parallel to the wide dimension. Except for the lower four inches, there shall be holes drilled through the center of the piece at one inch intervals to within approximately one inch from the top. The holes shall be proper clearance size for 3/16 inch stove bolts. The holes shall be drilled accurately so that they will line up properly for easy assembly and adjustment of the footpiece and bows. The lower three inches of the footpiece shall be rounded and reduced in size to a 7/8 inch diameter, tapering to a 3/4 inch ± 1/32 inch diameter at the end.

3.3 Assembly. The tenon of each bow shall be firmly anchored with small nails, pegs, or screws into the mortise in the arm or supporting piece. The handpiece shall be secured in position between the side bows with stove bolt, washers, and a wing nut. The lower ends of the two side bows shall converge with the footpiece between them and be secured with stove bolts, washers, and wing nuts. Each crutch shall be assembled prior to placing in unit package, so that the overall length of the assembled crutch shall be the minimum.

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3.4 Finish. Each wooden piece of each crutch shall have a protective coat of a good grade, clear, transparent spar varnish, which shall be properly applied before assembly and present a uniform hard, smooth surface and be free from bubbles and tackiness. Care shall be taken to prevent excess varnish from blocking clearance holes for stove bolts. If necessary, holes shall be reamed after finishing. A waterproof sealer coat shall be applied prior to the application of the varnish.

3.5 Strength. Crutches shall not break or otherwise fail when subjected to the tests prescribed in 4.4.1.

3.6 Identification marking. Each crutch shall be permanently marked on the inside top of the bow with the name of the manufacturer, contract number, and the lot number. Marking shall be by stamping or stenciling prior to the application of the varnish coat. Stencil ink or paint used shall be specific for the purpose. Marking shall be legible non-fading and durable. Color shall be black. As an alternate, the identification marking shall be placed on the side of the armpiece.

3.7 Workmanship. The workmanship shall be first class throughout. Crutches shall be free from any defects which may impair their serviceability, appearance, or safety.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Records. Records of examinations, and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, inspection level S-2 for AQL specified in 4.3.1.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

Table I. Sampling for tests

Component	Characteristic	Reqmt.	Test procedure	Unit of product	Insp. level	AQL (Percent defective)
End item	Strength	3.6	4.4.1	One individual crutch	S-2*	**
Wood	Moisture content	3.2.1	4.4.2	***	- -	***
	Shear strength (dry and wet)	3.2.1.2	4.4.3	***	- -	***

* A minimum of three units shall be tested, regardless of lot size.
** Acceptance number shall be zero for all sample sizes.
*** Each lot of wood shall be tested as specified in applicable test method. Failure to comply shall be cause for rejection of the lot represented.

4.2.3 Samples. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The crutches shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with, but not restricted to, the following classification of defects. Additional defects shall be classified in accordance with MIL-STD-105.

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Table II. Categories and defects

Major A		AQL 1.0
101	Wood components not free from warping, splits, splinters, decay, or checks.	
102	Surfaces of wood members not smoothly finished.	
103	Edges and corners of wood members not smoothly rounded.	
104	Armrest joints not securely fitted and properly assembled.	
105	Bolts, nuts, and washers on crutch fail to fit or assemble properly.	
106	Bolts, nuts, and washers not properly protected against corrosion.	
107	Nails protruding on armpiece.	
Major B		AQL 4.0
151	Handle and hardware fail to fit properly into bows in each adjustable position.	
152	Footpiece and hardware fail to fit properly into bows in each adjustable position.	
Minor A		AQL 2.5
201	Varnish finish omitted or not uniform and free from tackiness and bubbles.	
202	Footpiece stuck to bow.	
Minor B		AQL 4.0
231	Any dimensions not within tolerances specified.	
Minor C		AQL 6.5
261	Packaging, packing, and marking not as specified.	

A. Possible defects listed indicates minimum inspection requirements.
B. All dimensions are in inches unless otherwise specified.
C. Manufacturers (subcontractors and, or prime contractors) protocols for components and, or end items may be accepted by the Government in lieu of retesting
D. Unless otherwise specified, AQL indicates percent defective.
E. Unit of product for inspection purposes shall be the individual crutch, unless otherwise indicated.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Strength. Each sample crutch shall be tested as follows: Secure the crutch in a vertical position and apply a force of 300 pounds, in a vertically downward direction, to the center of the top surface of the armpiece. Remove this force and apply a vertical downward force of 150 pounds to the center of the handpiece. The crutch shall be secured or clamped for tests at any point or points not more than four inches below the top or above the bottom in such a manner that the bows will be completely unrestrained.

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4.4.2 Moisture content. The moisture content of the wood used in the fabrication of the crutches shall be determined as described in MIL-W-6110.

4.4.3 Shear strength (dry and wet). Test for dry shear strength shall be conducted in accordance with ASTM Designation D805-63, sections 31 through 35. Test for wet shear strength shall be conducted in accordance with ASTM Designation D805-63, sections 31 through 35, except that in this procedure, the test specimens shall be soaked for four hours in water at room temperature and then shall be tested immediately thereafter, while wet.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One pair of crutches, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. A sheet of kraft paper shall be interposed between the adjacent faces of each pair of crutches. Each pair of crutches (one unit) shall be packaged in a heat sealed polyethylene bag, 2.5 mils thick, and further packaged in a box of appropriate size and design, constructed in accordance with PPP-B-636, type CF, class domestic, fabricated from not less than 125 pound test fiberboard. Closure shall be adequate to prevent accidental opening.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.2.2 Level A.

5.2.2.1 Exterior container. Twelve units shall be packed in an exterior container designed for type 2 load and constructed in accordance with PPP-B-636, type CF, class weather-resistant, variety DW. Closure, strapping and waterproofing shall be as specified in the appendix of PPP-B-636. Boxes shall conform to the special requirements specified in Table II of PPP-B-636.

5.2.3 Level B.

5.2.3.1 Exterior container. Twelve units shall be packed in an exterior container designed for type 2 load and constructed in accordance with PPP-B-636, type CF, class domestic, variety DW. Method II closure specified in PPP-B-636 shall apply. Boxes shall conform to the special requirements specified in Table I of PPP-B-636.

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5.2.4 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with FED-STD-123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The crutch is intended for use as an aid in ambulation. It is designed for short term usage of hospital patients and convalescents as well as permanently disabled persons.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of crutch required (see 1.2).
(c) Selection of applicable levels of packing (see 5.1 and 5.2).

6.3 The crutch covered by this specification is listed in the Federal Supply Catalog under the following stock number:

National Stock Number	Item Identification
6515-00-777-7325	CRUTCH, WOOD, ADJUSTABLE

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the items which are normally procured to meet Federal Government requirements.

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